

Shubham Dattatray Asole

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Skills

Languages: Java, JavaScript, Python, SQL, TypeScript, Bash

Backend & APIs: REST APIs, Node.js, Express.js, Microservices, Distributed Systems, JWT, WebSockets

Security & Auth: JWT, RBAC Middleware, Secure Session Handling, Passport.js, Clerk Auth, Webhook Syncing

Frontend: React, Context API, Material UI, Chart.js, TypeScript

Databases: MongoDB, MySQL, PostgreSQL, Mongoose, Prisma ORM

AI & Integrations: Google Gemini API, LLM-based pipelines, Agent-based automation, REST webhook integration

DevOps & Cloud: Git, GitHub, CI/CD Basics, Linux/Unix, Vercel, Render, Sentry, Docker (awareness), AWS (learning)

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, System Design

Education

BTech in Electronics and Communication Engineering

Nov 2022 – May 2026

Indian Institute of Information Technology Design and Manufacturing, Kancheepuram

Key Projects

CLIPFORGE — AI Content Generation SaaS | React, TypeScript, Node.js, Express.js, PostgreSQL, Prisma

[GitHub](#) [Live](#)

- **Architected** a production-ready, full-stack **SaaS** platform on the **PERN** stack with a **Google Gemini API**-powered agent pipeline to automate viral marketing creative generation, reducing manual design effort to zero.
- **Engineered** a modular **RESTful** backend using **Node.js**, **Express.js**, **TypeScript**, and **Prisma ORM** to manage media assets, generation history, and user data across a scalable **PostgreSQL** database with robust API performance.
- **Secured** backend routes and user sessions by integrating **Clerk Auth** with custom **RBAC middleware** and **webhook**-based data syncing, establishing a production-grade access control architecture aligned with **OOP**-driven design principles.
- **Implemented** a credit-based usage-limit system to enforce a scalable subscription monetization model, enabling per-user API quota tracking and dynamic plan enforcement across all backend routes.

LoadMitrra — Full-Stack MERN Freight Matching & Logistics Platform

[GitHub](#) [Live](#)

- **Designed** and implemented custom **RBAC middleware** using stateless **JWT** authentication and secure **Axios** interceptors to enforce backend security across all **REST API** routes, eliminating unauthorized cross-portal access.
- **Built** a high-throughput **REST API** layer using **Express.js** with optimized nested **Mongoose** schemas to enforce data consistency across a complex 5-stage shipment lifecycle supporting dynamic state transitions and automated billing.
- **Replaced** HTTP polling with a real-time, room-based **WebSocket** communication layer via **Socket.io**, achieving near-zero message latency for cross-user load coordination.
- **Integrated** **React Google Maps SDK** and **Leaflet APIs** for coordinate-driven visual rendering, improving location-based load discovery and delivery route accuracy for dual user bases.
- **Deployed** the full-stack platform on **Vercel** and **Render** with optimistic UI rendering fallbacks, ensuring 100% perceived message delivery success in low-connectivity environments.

StatzHub — Full-Stack Trading & Investment Platform | MERN Stack

[GitHub](#)

- **Developed** modular **REST APIs** using **Node.js** and **Express.js** to manage users, orders, holdings, and positions with scalable **CRUD** operations and structured routing across a multi-app architecture.
- **Implemented** secure authentication using **JWT** with **OOP**-aligned, middleware-based route protection, enabling authorized access flows across a marketing site and a live trading dashboard.
- **Designed** optimized **MongoDB (Mongoose)** schemas with validations and entity relationships to ensure consistent, reliable **NoSQL** storage for trading data including users, orders, positions, and portfolios.
- **Integrated** portfolio analytics using **Chart.js** to visualize performance distribution through interactive doughnut and bar charts, improving data-driven decision support for end users.

Professional Experience

Embedded Systems Intern — Startrit Infratech Pvt Ltd

May 2025 – July 2025

- **Developed** and integrated sensor-based control modules using **Embedded C** and **Python** to improve automation system reliability and support stable real-time operations.
- **Validated** hardware-software communication by building and testing a **Raspberry Pi** robotic prototype over **UART**, ensuring accurate data transmission and device control.
- **Accelerated** inspection workflows by executing **MATLAB**-based simulations and performing structured data analysis to support engineering decisions.

Certifications

Programming in Java — NPTEL (IIT Certified, 12-week)

Jan – Apr 2024

Score: 25/25 (Assignment) | 67% overall | 3–4 Credits | ID: [NPTEL24CS43S1053409139](#)

Database Management System — NPTEL (IIT Certified, 8-week)

Jul – Sep 2025

Score: 21.67/25 (Assignment) | 61% overall | ID: [NPTEL25CS145S633209887](#)